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CIS Senior Cybersecurity Project's proposal to fulfil the requirement of the CIS-Cybersecurity degree

# Secure Peer-to-Peer Messaging Using Distributed Servers

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## Abstract

This project designs, implements and demonstrates a secure, highly available peer-to-peer messaging system. Peer to peer communications leverage WebRTC for end-to-end encrypted communication channels. It is supported by a distributed mesh of servers for peer discovery and for WebRTC channels with TURN relays. The system will be decentralized, making it more resistant to targeted availability attacks. Peer to peer connections will be used to minimize extra metadata that could be used in a metadata attack. Security phrases will be displayed so users can validate the security of their communication channel. This project aims to demonstrate the practicality of using a decentralized model for secure communications.

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# Introduction

## Project Overview

This project explores the design and implementation of a highly available, peer-to-peer messaging application. It is built on top of a mesh of distributed light weight servers that are used to facilitate the peer-to-peer connection between clients. Instead of using a single centralized server for coordinating and providing a source of truth, the system will use an algorithm to share, find and route peer discovery. This algorithm will be similar to how OSPF shares routes between routers, except it will be sharing peers between servers. The system will leverage WebRTC for the peer-to-peer with built in end-to-end encryption for message sending. The decentralized server mesh will be used for connection negotiation and network traversal. Unlike a traditional centralized messaging platform, this architecture avoids single points of failure, allowing for normal operations even if a few servers become unresponsive.

## Context & Relevance

Traditional messaging platforms are more reliant on centralized models which make them particularly suspectable to targeted availability attacks. This includes targeted denial of server attacks, server comprises, physical breaches, data center outages and more. An attack which successfully takes down a centralized server could cause serious downtime, or severe service degradation. In this decentralized system, many of these decentralized servers would need to be taken down all at once before any service degradation could occur. Virtually every messaging platform is susceptible to meta-data attacks. While the messages may be end-to-end encrypted, the service provider, or an attack which gains control of the service provider can still see who sent who messages and when. The peer-to-peer messaging approach helps to reduce this attack surface. This project is relevant because it directly relates to the fields of computer networking, cybersecurity and distributed systems.

## Goals & Objectives

The primary goal of this project is to design, implement and demonstrate a distributed server mesh that enables secure, reliable peer-to-peer communication without reliance on a single

centralized service (Note: DNS or domain name will be centralized). This includes a self-healing mesh of servers which can share information between themselves to facilitate a peer-to-peer connection, implementing a WebRTC web application with end-to-end encrypted communication, and demonstrating the functionality of the system even during a targeted availability attack, its ability to mitigate a man-in-the-middle attack, and it's lower profile metadata trail to avoid metadata attacks.

### Significance

This project is significant because it is designed to be distributed from the ground up. It showcases how projects designed with a distributed architecture in mind, rather than adding it on later, can have many cyber security related benefits, besides just better performance.

## Background & Related Work

### Literature Review

Using a mesh system with peer-to-peer communication isn't a new concept. This has been previously implemented and it critical for resisting censorship and failing infrastructure (Patel et al., 2025). Distributed hash tables can provide a O(log n) lookup efficiency, but are subject to attacks which can corrupt data or destroy its validity (Patel et al., 2025)

WebRTC can provide low latency browser-based P2P communication with core APIs (WebRTC, 2019). STUN can be used for NAT traversal with a 92& success rate and TURN as a fallback relay 8% of the time (Mahmoud & Abozariba, 2024). Security is already built into WebRTC which ensures secure end-to-end communication.

There are many existing security protocols, especially for end-to-end encrypted messages. Signal famously created their Double Ratchet algorithm to provide forward secrecy and post compromise security (Bienstock et al., 2023). However, Signal's Double Ratchet does nothing to protect against meta data exploits. In this project, the Double Ratchet will not be used to avoid this metadata exploitation.

### Research Gaps

WebRTC is primarily used for audio and video transmission. As a result, there is minimal research done to use it for a messaging application, in contrast. Much of the research that does use it for non-audio-visual purposes is limited in scope and scalability. Research also does not often demonstrate the self-healing abilities of distributed systems, despite that being a common trait.

### Related Work

The ASMesh protocol is likely the closet example to what this project aims to accomplish. It is more complex in many ways. It uses Signal's Double Ratchet encryption and focuses on extending research on cryptography. (Bienstock et al., 2023)

There are also many existing WebRTC applications with a proven track record. However the majority of these WebRTC applications still rely on a single centralized server which results in a single point of failure (Mahmoud & Abozariba, 2024)

Many current messaging systems adopt a federated architecture. Federation enables traditional centralized services to be split across multiple servers to improve scalability and availability. In many ways this is similar to what is currently being implemented. The servers will be distributed like with federated communication, but ultimately this project will be more decentralized than federated. This is because this design doesn't designate a precipitant authority and single server in the mesh will ever maintain a persistent identity (Li et al., 2025). Instead, all servers will be treated as equals and all clients treated as equals.

## Problem Statement

### Problem Description

Traditional messaging platforms relay on centralized servers which makes them highly susceptible to targeted availability attacks like DDoS. This can result in severe service degradation and or a complete outage. Messaging systems can also be susceptible to man-in-the-

middle attacks, even if they are end-to-end encrypted if they do not provide a security phrase. Finally, every mainstream messaging application suffers from metadata attacks. It's easy to see who is talking to who and when. Using a distributed mesh of severs and peer-to-peer direct client communication, this project aims to reduce the attack surfaces of each prior mentioned attack.

**Scope of the Problem**

This project is limited to just live communication; there will be no mailbox. Both users must be active to receive messages. This project will not entirely eliminate availability attacks, as a distributed DDoS, complete outage of all servers, or internet outage will still cause an outage for users. This project will be unable to fully mitigate metadata attacks because the servers will still be aware of who is who talking to who, just not how often and how much. Finally, group chats will not be supported due to the exponential complexity with end-to-end encryption and peer to peer networks.

**Challenges**

There are many challenges that this project faces. First, distributed systems are much more difficult to develop and debug problems. Peer to peer networks can also be unreliable due to strict network requirements. WebRTC aims to make this much more approachable, especially when combined with TURN.

**Social Impact**

Having a highly available and minimally tracking messaging platform is essential to the freedom of speech and the fight against censorship. The proof of concept of this project could be turned into a commercial product which could enable users to still communicate. This communication can still occur even if the system is being attacked by adversary due to its highly distributed design.

### Risk Management

It's possible that WebRTC will not be able to reliably create a connection between peers. In this case, there is no good backup option besides readdressing the scope of this project. However, WebRTC is a reliable standard which has been mainstream for many years now.

### Ethics

This project will use open-source software to avoid any ethical concerns or misuse of paid software products. The software may include, but is not limited to Golang, JavaScript, TypeScript, Chromium, Visual Studio Code, Docker, Vite.

## Proposed System and Expected Results

### Proposed Solution

This project aims to create a highly distributed mesh of servers to help facilitate peer to peer connections which will enable end-to-end encrypted messaging. The highly distributed nature of the servers makes targeted availability attacks much less effective. The peer-to-peer connections help to mitigate metadata attacks by eliminating excess data being sent through the servers. Finally, man-in-the-middle attacks is avoided by displaying a security phrase which can be validated via a separate channel.

### Methodology

The distributed mesh of servers will be implemented in a programing language apt to server development, such as Golang. Because of the distributed nature of the servers, a containerization environment like Docker may be used to help assist in multiple deployments of identical images. The client application will be a web application using JavaScript or possibly TypeScript to interact with WebRTC, the distributed servers and the user. When the system is completed, the availability attacks will be tested by shutting down servers and ensuring availability remains. Metadata attacks can be tested by monitoring generated traffic to show a

decrease in visible information outside of the client devices. Man-in-the-middle attacks can be tested by verifying the security phrase via a separate channel.

**Expected Results**

At the completion of this project, it should demonstrate a highly available distributed mesh of servers that facilitate a peer-to-peer connection for end-to-end encryption. This looks like a user being able to load a webpage, create a messaging session with another active user and continue to communicate and create new sessions even has servers are turned off and on unpredictably. Users will be able to verify the security of their communications by checking for a security phrase which will guarantee the identity of the other user.

## References

Bienstock, A., Rösler, P., & Tang, Y. (2023, September 5). *ASMESH: Anonymous and secure messaging in mesh networks using stronger, anonymous double ratchet*. IACR Cryptology ePrint Archive. https://eprint.iacr.org/2023/1053

*Getting started with WebRTC*. WebRTC. (2019, March 28). https://webrtc.org/getting-started/overview

Mahmoud, H., & Abozariba, R. (2024, November 23). *A systematic review on WebRTC for potential applications and challenges beyond audio video streaming - multimedia tools and applications*. SpringerLink. https://link.springer.com/article/10.1007/s11042-024-20448-9

Patel, R., Biswas, U., Kodipaka, S., Carroll, W., Peranich, P., & Young, M. (2025, September 23). *A Survey of Recent Advancements in Secure Peer-to-Peer Networks*. https://arxiv.org/. https://arxiv.org/abs/2509.19539